

Group 2: Ben, Drew, Gabe, Joseph

Objective

This project is about deeply understanding the material, connecting concepts across the semester, and demonstrating ownership of your learning. You'll revisit the two core papers, apply the methods with MATLAB, and reflect clearly on the process and your learning.

Part 1: Understand and Explain the Methods

Paper 1: Signal Estimation Using RBF

Explain how Radial Basis Functions (RBFs) estimate signal values at unmeasured directions (● blue). Include diagrams of the setup.

Break down the mathematical idea behind the method in clear, accessible language.

Paper 2: Constructing ODFs

Explain what an Orientation Distribution Function (ODF) is and why it's important.

Describe how signals are used to construct ODFs.

Include a visual explanation (diagram or figure) showing the process.

Part 2: MATLAB Application

- Use the provided MATLAB code (no new code needed).
- Apply the RBF method to the class dataset.
- Use the result to compute and visualize an ODF.
- Add comments to the code or explain the steps clearly in your report.

Part 3: Reflection and Synthesis

- Summarize the workflow from signal $\rightarrow$ ODF.
- Reflect on what you now understand more deeply.
- Describe what challenged you and how you worked through it.
- Explain the broader significance or application of this process.

Submit the following:

- MATLAB script(s) or function(s)
- Presentation slides (PDF or PPT)
- Report 4-6 pages in LaTeX.

Grading Criteria (100 points total)

Category	Points	What I'm Looking For
Understanding Paper 1	20 pts	Clear explanation, diagram, and understanding of estimation
Understanding Paper 2	20 pts	Explanation of how ODFs are constructed from signals
MATLAB Application	20 pts	MATLAB code used correctly, results shown and interpreted
Presentation	20 pts	Clear, well-structured, engaging; 30-40 minutes.
Written Report	15 pts	Concise, informative, and well-organized report explaining process and findings
Ownership & Effort	5 pts	Graded by audience: Did the presenter show care, effort, and genuine engagement?